

Wavelength and Polarization Diversity OTDR for Highly Accurate Fiber Faults Identification

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Abstract We experimentally demonstrate a wavelength- and polarization-diversity OTDR for fiber faults identification in data-center networks. Our system accurately distinguishes fiber faults beyond conventional OTDR capabilities. Six critical fault types are emulated, and the proposed OTDR architecture, aided by ML, demonstrates over 99% faults identification accuracy. ©2026 The Author(s)

Introduction

In the era of rapidly expanding AI-driven applications, modern data centers impose stringent requirements on the bandwidth, latency, and reliability of data transmission systems [1]. Fiber-optic links interconnecting switches, servers, and storage systems form the backbone of high-speed communication and are crucial elements for meeting these demands. However, optical fiber links become increasingly vulnerable to physical impairments and faults as data rates continue to scale [2-4]. Inter-data-center links' faults have been reported to last 1.27 hours on average and exceed 3 hours in the tail case, often leading to service disruptions [5]. According to Federal Communications Commission (FCC), more than one-third of service disruptions are attributed to fiber-related issues [6]. Consequently, comprehensive fiber testing is essential for ensuring high-reliability optical transmission [7].

Optical time-domain reflectometry (OTDR) is widely used for fiber fault management by exploiting reflections and Rayleigh backscattering [8]. Accurately identifying the underlying fault types from OTDR traces is critical for effective fault diagnosis and network maintenance [6]. Leveraging the powerful features extraction and pattern recognition capabilities of machine learning (ML) models, Abdelli et al. [9] proposed a data-driven framework combining an autoencoder and an attention-based bidirectional gated recurrent unit (BiGRU), achieving high accuracy in detecting and identifying faults such as fiber tapping, bad splice, dirty connector, and fiber cut. Straub et al. [10] proposed a random forest-based OTDR classification framework, achieving 98% precision and 95% recall on measured traces for reflection, attenuation, and no-event categories.

Despite these promising results, the existing approaches primarily rely on the shape of OTDR traces for fault discrimination, limiting their ability to distinguish faults that exhibit similar signatures under single-wavelength and single-polarization measurement scenarios. In practice, optical fiber

links also suffer from other critical impairments, including microbending, macrobending, loose connectors, and connector mismatch [11], which introduce attenuation and reflection, degrading signal integrity. Moreover, some of these fiber faults produce highly similar OTDR trace patterns. For example, bad splices and bending-induced losses often exhibit comparable attenuation features, making them difficult to distinguish using conventional OTDR with a single wavelength and polarization [12]. Moreover, existing approaches largely overlook the underlying physical mechanisms of fiber faults. In fact, different fault types exhibit distinct responses under wavelength and polarization variations, as shown in Tab. 1. Systematically probing these responses can yield high-dimensional representations with enhanced discriminative capabilities for faults identification.

In this paper, we propose a wavelength and polarization diversity OTDR (WPD-OTDR) framework for accurate fiber faults identification. By scanning the fiber using 3 wavelengths and 11 polarization states, the approach constructs high-dimensional signal representations that capture the unique intrinsic responses of 6 major fiber fault types. Experimental results demonstrate that our method can effectively distinguish various fault types that exhibit highly-similar OTDR traces under conventional measurements, achieving a 99.05% identification accuracy.

Tab. 1: Wavelength & polarization dependence of fiber faults.

Fiber fault type	Wavelength dependence	Polarization dependence
Macrobending	Strong [13]	Strong [14]
Microbending	Strong [15]	Strong [16]
Bad splice	Moderate [17]	Weak [18]
Connector contamination	Moderate [19]	Weak
Loose connector	Weak [20]	Moderate [21]
Connector mismatch	Weak	Moderate [21]

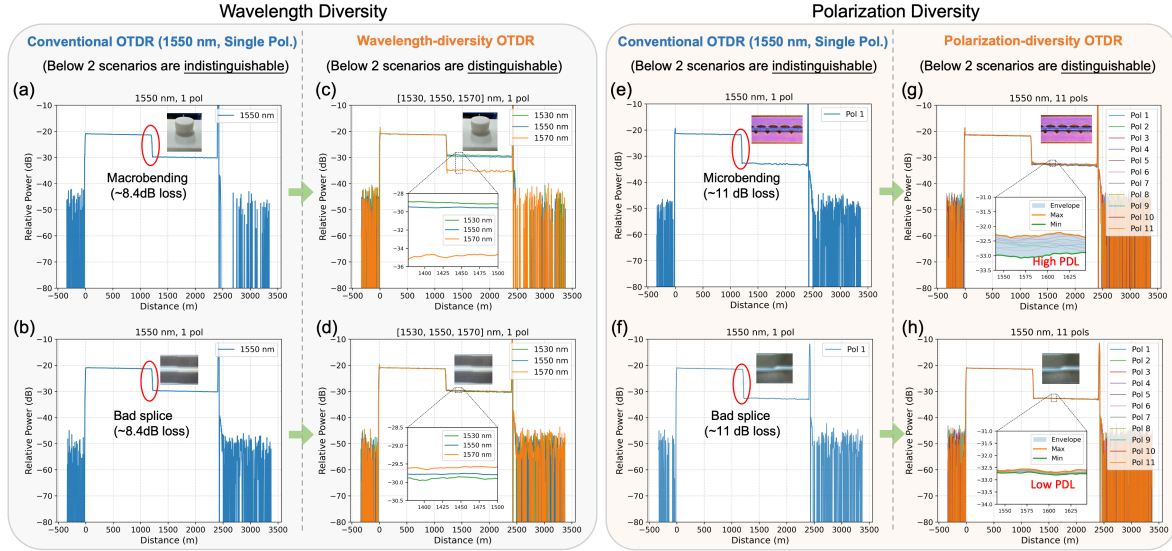


Fig. 1: Illustration of enhanced fiber faults discriminability using WPD-OTDR. (a)-(b) Conventional OTDR, where two considered fault types produce indistinguishable traces. (c)-(d) Wavelength-diversity OTDR, enabling faults separation. (e)-(f) Conventional OTDR. (g)-(h) Polarization-diversity OTDR, enabling improved faults discrimination.

Operating Principle

Conventional OTDR is limited in distinguishing fiber faults with similar trace shapes under single-wavelength and polarization-insensitive measurements. Such ambiguities are particularly evident in macrobending, microbending, and bad splicing scenarios, which produce nearly indistinguishable OTDR traces. Specifically, as shown in Fig. 1(a) and (b), macrobending and a bad splice (both causing 8.4 dB loss) produce nearly identical OTDR traces under conventional measurements. However, these two fault types exhibit fundamentally different wavelength dependencies. That is, macrobending loss is highly dependent on wavelength, with higher attenuations occurring at longer wavelengths [13]. In contrast, bad splice-induced loss due to lateral misalignment is governed by mode-field mismatch, where increasing the wavelength expands the mode field and improves modal overlap, thereby reducing the loss and leading to a wavelength dependence that is opposite to that of macrobending case [17]. As a result, the two faults become clearly distinguishable when the wavelength diversity is introduced, as shown in Fig. 1(c) and (d). Similarly, as shown in Fig. 1(e) and (f), microbending and a bad splice produce nearly identical

OTDR traces under conventional measurements. However, microbending exhibits strong polarization-dependent loss (PDL) due to stress-induced birefringence, resulting in unequal attenuation for different polarization states [16], whereas bad splice-induced loss is relatively polarization-independent [18], enabling clear faults separation when polarization diversity is introduced, as shown in Fig. 1(g) and (h). Overall, by exploiting wavelength and polarization diversity, fault discriminability is significantly improved, forming the basis of the proposed fiber faults identification framework.

Proposed Framework Implementation

Schematic diagram of our implemented WPD-OTDR is illustrated in Fig. 2. Three pulsed laser sources with center wavelengths of 1530 nm, 1550 nm, and 1570 nm were employed, and a programmable polarization controller (PC) was used to generate multiple polarization states. The generated pulses were then launched into standard single-mode fiber links of various lengths. The backscattered signals were collected via a circulator and detected using an avalanche photodiode (APD), followed by their acquisition and averaging to obtain OTDR traces. For each

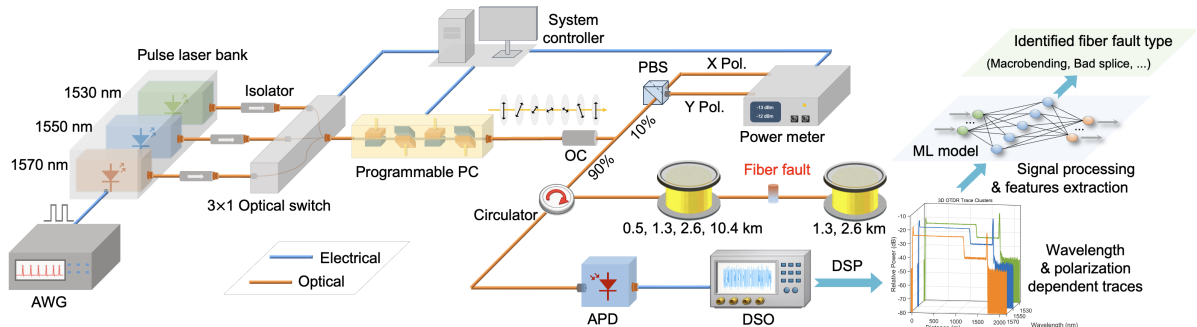


Fig. 2: Schematic diagram of the implemented wavelength- and polarization- diversity OTDR for identifying fiber faults.

emulated fault instance, measurements were conducted at 3 wavelengths, each with a scan over 11 polarization states, resulting in 33 OTDR traces per fault instance. Prior to each measurement, 10% of optical power at the PC output was tapped for monitoring and used for polarization initialization to ensure consistent starting states.

Next, from the collected multi-dimensional OTDR traces, both physical and morphological features were extracted for fault identification. Specifically, for each fault instance, 33 traces were first processed using a sliding window-based event detection method to isolate local fault regions. From these detected events, relative wavelength-dependent loss and PDL variations were computed to capture the intrinsic physical responses of different faults, forming a set of 6-dimensional physical features. In addition, to characterize the shape of the OTDR traces while mitigating the influence of absolute loss values, we applied differential and normalization operations to each trace, yielding a set of 30-dimensional morphology-based features. The physical and morphological features were then concatenated into a 36-dimensional feature vector for each fault instance. These feature vectors were fed into a feedforward artificial neural network (ANN) with two hidden layers (64 and 32 neurons, respectively) and an output layer corresponding to six fault classes. A dataset comprising six fault categories, as defined in Tab. 1, with 29, 27, 50, 39, 30, and 35 data samples, respectively, was collected under diverse experimental conditions. Based on this dataset, the model was trained and evaluated using five-fold cross-validation.

Results and Discussion

We first analyzed the discriminability of different fault types using conventional OTDR measurements. Fig. 3(a) presents the linear discriminant analysis (LDA) visualization [22] based on trace-shape features obtained from single-wavelength and polarization-insensitive measurements. It can be observed that macrobending, microbending, and bad splice exhibit significant overlap due to their similar trace shapes, making them difficult to distinguish, whereas the remaining fault types are relatively well separated owing to more distinct morphological characteristics. This is also reflected in the results in Fig. 3(b), where conventional OTDR achieves 72.86% overall accuracy but below 50% for the above-mentioned three fault types. In contrast, the proposed WPD-OTDR significantly improves fault discriminability. As evident from Fig. 3(c), the previously overlapping fault types become clearly separable, leading to a substantial improvement in classification performance, with an overall accuracy of 99.05% as shown in Fig. 3(d). These results confirm the effectiveness of the proposed method in overcoming the limitations of conventional OTDR devices.

Conclusions

We proposed a novel WPD-OTDR framework that resolves the ambiguities of conventional OTDR in distinguishing fiber faults with similar trace patterns. Experimental results demonstrated an accuracy of 99.05%, highlighting its great potential for intelligent and automated optical fiber links monitoring in future high-capacity data-center networks.

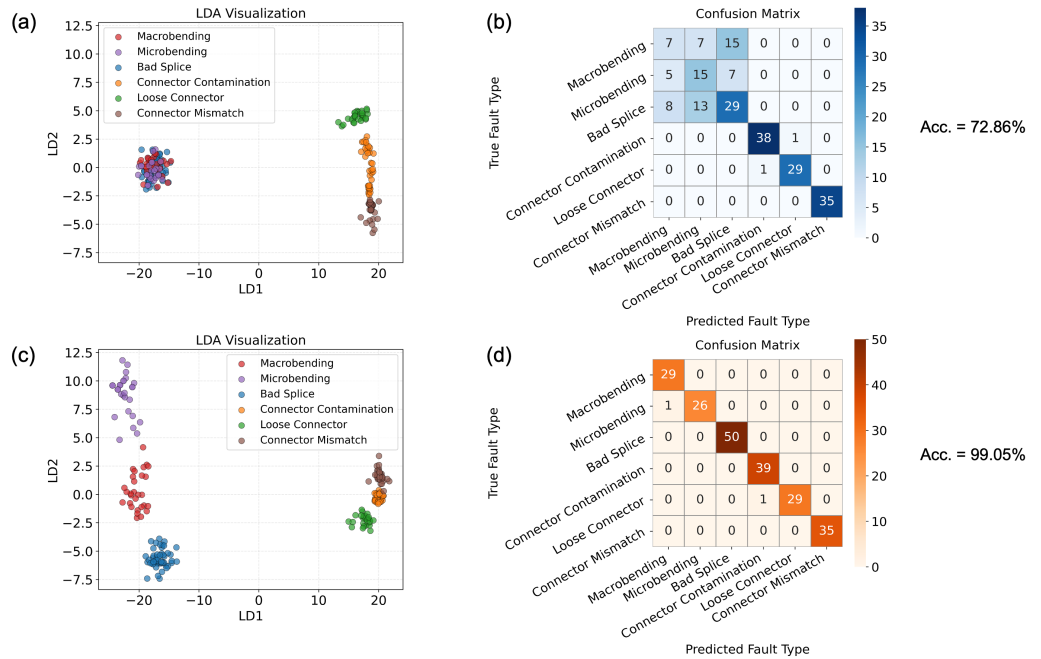


Fig. 3: Comparison of fiber faults discriminability using conventional OTDR and the proposed method. (a) LDA visualization, and (b) corresponding confusion matrix using conventional OTDR. (c) LDA visualization, and (d) corresponding confusion matrix using the proposed method.

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